THE UNITED REPUBLIC OF TANZANIA

PRESIDENT'S OFFICE

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

NORTH WESTERN ZONE FORM FOUR FORM PRE-NATIONAL EXAMINATION

032/1 TIME: 3HRS CHEMISTRY - 20/9/2024

INSTRUCTION.

1. This paper consists of section A, B and C with a total of eleven (11) questions.
2. Answer all question in sections A and B and two (2) questions from section C.
3. Non – programmable calculators may be allowed.
4. Write your examination Number on every page of your answer sheet (s).
5. The following constants may be used.

   Atomic masses:

   Atomic masses H=1, O=16, Na=23.

   C = 12, Cl=35.5, N=14, Zn=65

   S=32, Ca=40, Fe=56, Cu=64

   Al=27.

   GMV at S.t.p = $22.4dm^3$

   1 faraday = 96500 coulombs

   Avogadro's number = 6.02×10^{23}

   1 litre = $1dm^3=1000cm^3$

SECTION A (16 Marks)

Answer all questions in this section.

1. For each of the following, items (i) – (x) choose the correct answer among the given alternatives and write its letter beside the item number in the answer sheet (s) provided.

   i) Splints are among the first aid kit components which of the following sentences best describes it use?

      a) Used for cutting bandages or piece of cloth.
      b) Used for holding pieces of bandages or cloth together
      c) Used to support broken bones and is tied using bandages
      d) Used for picking up and holding things
      e) Used for cleaning wounds and make it heal faster

   ii) A simple proof that some chemical reactions take place in our bodies is that

      a) We eat a balanced diet
      b) Doctors tell us so in the hospitals
      c) We occasionally fall sick
      d) The food we eat or the drinks we take are quite different from the waste products from our bodies
      e) There is no proof.

   iii) The following reaction, $4NH_{3(g)}+50_{2(g)} \rightarrow 4NO_{(g)}+6H_2O$ (g), is an example of a

      (a) Decomposition reaction
      (b) Combination reaction
      (c) Redox reaction
      (d) Neutralization reaction
      (e) Esterification reaction

   iv) An aqueous solution turns red litmus solution blue. Excess addition of which of the following solution would reverse the change?

      (a) Baking powder
      (b) Lime
      (c) Aqueous ammonia
      (d) Vinegar
      (e) Quick lime

   v) Oils on treating with hydrogen in the presence of nickel catalyst from fats. This is an example of –

(a) Oxidation reaction

(b) Addition reaction

(c) Displacement reaction

(d) Substitution reaction

(e) Precipitation reaction

vi) Three elements T, Q and R, are in the same period of the periodic table. the oxide of T is amphoteric, the oxide of Q is basic and the oxide of R is acidic. Which of the following shows the elements arranged in order of increasing atomic number?

(a) T, Q, R

(b) Q, T, R

(c) Q, R, T

(d) R, T, Q

(e) T, R, Q

vii) What happens when dilute sulphuric acid is poured on copper plate?

(a) Zinc sulphate formed

(b) Zinc chloride formed

(c) Copper chloride formed

(d) None of the above

viii) The oxidation state of chlorine in potassium chlorate is:-

(a) +1

(b) +7

(c) +5

(d) -1

(e) +6

ix) The empirical formula of a certain compound is CH_3. Its molar mass is 30g/mol. what will be its molecular formula?

(a) C_2H_6

(b) CH_4

(c) C_2H_8

(d) C_2H_4

(e) C_3H_8

x) Ten gram of copper powder were placed in a sealed initiation tube containing air. the tube was heated very strongly for half an hour. At the end of the experiment

(a) Water vapor appeared in the tube

(b) Undergoes decomposition

(c) The total mass of the tube and its contents decreased

(d) Copper melted into a liquid copper form

(e) Copper increased in mass and changed in colour.

2. Match the items in List A with the correct responses in List B

LIST A	**LIST B**
(i) Methyl orange indicator	A. Normal salt
(ii) Calcium hydroxide	B. Concentrated base
(iii) PH_2	C. $H^+_{(aq)} + OH^-_{(aq)} \longrightarrow H_2O_{(L)}$
(iv) Neutralization reaction.	D. Composition reaction
(v) Sodium hydrogen Sulphate	E. Strong acid + weak base
(vi) Anode	F. Slaked lime
	G. Strong base + weak acid
	H. Acidic salt
	I. Concentrated acid
	J. Neutral salt
	K .Positive electrode
	L. Negative electrons

3. You are provided with the following fuels fire woods, natural gas, charcoal and kerosene.
   (a) Giving three reasons, explain why natural gas is the best option among the fuels given above for domestic purpose. Four (4) points
   (b) Giving five (5) reasons, explain why fire wood and charcoal which are cheap and most available fuels are not recommended for domestic uses.

4. (a) (i) People suffering from heart burn usually use wood ashes for relief. Mention characteristics which make the ashes to be used for heart burn relief.
   (ii)Give four compounds found in the laboratory which show the same characteristic as ashes.
   (b)How many ions are there in 6.82g of $Al_2(SO_4)_3$.

5. A student tested four sample of water each 10cm3 different areas at Malawi village by shaking with tree drops of soap solution. The experiment was repeated a second time by boiling each sample of water ($10cm^3$) with three drops of soap solution. The observation was recorded as shown in the table below.

Sample	Observation with Soap solution	Observation for boiled sample with Soap
A	No lather	Lather
B	Lather	Lather
S	Lather	Lather
D	No lather	No lather

(a) Which sample contains only temporary hard water? Give reason.

(b) Which sample contains permanent hard water? Give reason

(c) Name the chemical substances that would be the causes of hardness in sample A.

(d) Write the chemical equation for removing hardness in

(i) Sample A

(ii) Sample D

6. 5.3g of X_2CO_3 was dissolved in water to make 0.5 litres of a solution. $25cm^3$ of this solution required $50.0cm^3$of 0.1MHCL for complete neutralization.

(a) Write the balanced chemical equation for the reaction

(b) Calculate the concentration of X_2CO_3 in mol/dm^3

(c) Calculate the relative molecular mass of X_2CO_3

(d) Calculate the relative atomic mass of X

(e) What is the name and symbol of element X

7. (a) A student was preparing food for the family using hot oil on a frying pan. Accidently, the pan tipped over and a huge fire spread on the kitchen floor.

(i) Which fire extinguishers would be suitable for putting out the fire? Explain

(ii) Which fire extinguishers would not be suitable for putting out the fire? Explain

(b) The following results were obtained in an experiment to find the molecular formula of magnesium oxide when magnesium burnt in air:-

Mass of crucible = 12. 6g

Mass of crucible + magnesium = 14.9g

Mass of crucible + magnesium oxide = 16.5g

(i) What mass of oxygen combined with magnesium?

(ii) What was the empirical formula of this oxide?

8 (a) Why does a mixture of iron and sulphur after heating starting to glow and continues to glow even after the external source of heat is removed?

(b). Write the balanced equation for the reaction that takes place in (a) above.

(c). pure nitric (V) acid is colourless liquid. Explain why the one prepared in the laboratory is yellow?

SECTION C (30 MARKS).

Answer only two (2) questions from this section

9. With the aid of the balanced chemical equation explain the processes that occur in the blast furnace during extraction of iron.

10. (a)Briefly explain the following observations with the help of equations

(i) White anhydrous copper (II) Sulphate changes it colour to blue when water is added.

(ii) Vigorous reaction takes place when a small piece of sodium metal is placed in water.

(iii) Addition of Zinc metal into a solution of copper (II) Sulphate result into decolonization of the solution and deposition of a brown solid substance.

(b)Define the following terms and give one example in each case

(i) Weak acid

(ii) Acidic - salt

11. We have coal at Kiwira in Mbeya region. Authorities in government have allowed use of the coal for domestic and industrial purpose. What warning can you rise concerning likely effects